SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:		**Duration:**	60 minutes

Code of Conduct

I K Y V Satyanarayana Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *K Y V Satyanarayana Reddy*

Annexure A – Architecture Design Assignment

Business Intelligence Dashboard for Executive Decision Support

1. Analyse System Requirements

1.1 System Objectives

The primary objective of the **Business Intelligence Dashboard for Executive Decision Support** is to provide executives and managers with a centralized platform for analysing organizational data and making informed business decisions. The system integrates data from multiple departments such as sales, finance, inventory, marketing, and human resources into a single dashboard. It provides real-time reports, interactive visualizations, and Key Performance Indicators (KPIs), enabling management to monitor business performance, identify trends, forecast future outcomes, and improve operational efficiency.

1.2 Functional Requirements

- User Registration and Login
- User Authentication and Authorization
- Dashboard Management
- Data Collection from Multiple Sources
- Data Integration
- Data Analysis
- KPI Monitoring
- Sales Analysis
- Financial Analysis
- Inventory Monitoring
- Employee Performance Tracking
- Customer Analytics
- Report Generation
- Interactive Charts and Graphs
- Data Filtering and Search

- Export Reports (PDF/Excel)
- Notification and Alerts
- User Management
- Audit Logging
- Database Backup and Recovery

1.3 Non-Functional Requirements

Performance

- Dashboard should load within 3 seconds.
- Reports should generate quickly.
- Support concurrent users efficiently.

Security

- Secure login authentication.
- Role-based access control.
- Data encryption.
- Secure API communication.

Reliability

- Accurate report generation.
- Automatic backup.
- Fault tolerance.

Usability

- User-friendly interface.
- Responsive dashboard.
- Easy navigation.

Scalability

- Support increasing users.
- Handle growing business data.
- Easy integration of new modules.

Maintainability

- Modular architecture.
- Easy software updates.
- Well-documented code.

Availability

- 24×7 system availability.
- Disaster recovery support.
- Minimum downtime.

1.4 Quality Attributes

Scalability

- Supports thousands of users.
- Easily expands with business growth.
- Supports cloud deployment.

Maintainability

- Modular design simplifies updates.
- Independent component maintenance.
- Easy debugging.

Reliability

- Automatic backup.
- Data validation.

- Error handling.

Availability

- High uptime.
- Load balancing.
- Backup servers.

Performance

- Fast dashboard loading.
- Optimized database queries.
- Data caching.

Security

- Multi-factor authentication.
- Data encryption.
- Secure APIs.
- Role-based permissions.

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2. Select a Suitable Software Architecture

Selected Architecture: Layered Architecture (Three-Tier Architecture)

Justification

The **Layered Architecture** is the most suitable architecture for the Business Intelligence Dashboard because it separates the system into independent layers such as Presentation Layer, Business Logic Layer, and Data Layer. This separation improves maintainability, scalability, and security while simplifying development and testing. Each layer performs a specific function and communicates only with adjacent layers, making the system easier to manage and extend. It also supports integration with external databases, APIs, and cloud services.

Architecture Layers

1. Presentation Layer

- User Interface
- Dashboard
- Reports
- Charts

2. Business Logic Layer

- Data Processing
- KPI Calculation
- Analytics Engine
- Report Generation

3. Data Access Layer

- Database Access
- Data Retrieval
- Data Storage

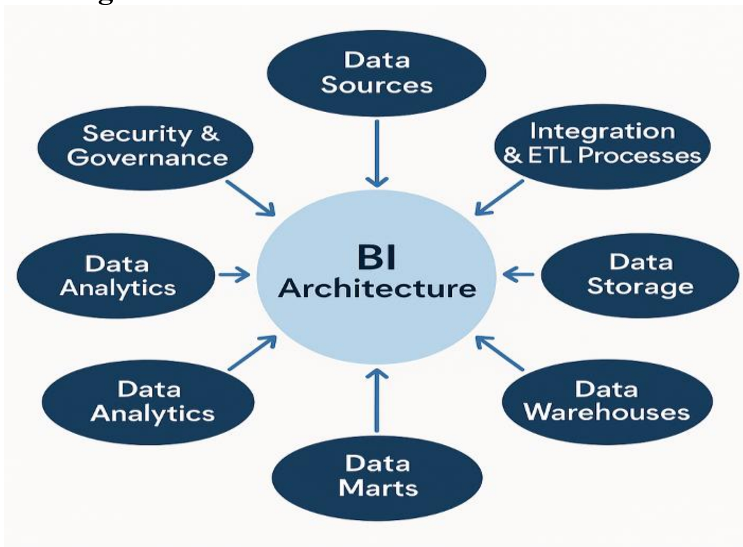
4. Database Layer

- SQL Database
- Data Warehouse

5. External Systems

- ERP System
- CRM System
- Cloud Storage
- APIs

3. Software Architecture Diagram



4. Explain Components and Their Interactions

4.1 Presentation Layer

The Presentation Layer provides the graphical user interface through which executives, managers, and analysts interact with the system. It displays dashboards, business reports, charts, graphs, and KPIs. Users can search data, apply filters, and generate reports using this layer.

4.2 Business Logic Layer

This layer performs all business operations such as data analysis, KPI calculation, report generation, forecasting, and business rule implementation. It processes incoming data and prepares meaningful insights before sending them to the presentation layer.

4.3 Data Access Layer

The Data Access Layer acts as a bridge between the business logic and the database. It retrieves, updates, inserts, and deletes data while ensuring secure database communication. It also integrates external APIs and data sources.

4.4 Database Layer

The Database Layer stores all organizational information such as employee records, customer data, sales information, financial reports, inventory details, and KPI data. It ensures data consistency, integrity, and secure storage.

4.5 External Systems

External systems include ERP, CRM, HRMS, cloud storage, and third-party APIs. These systems continuously provide updated business data to the dashboard, enabling real-time business intelligence and analytics.

4.6 Component Interaction

1. User logs into the dashboard.
2. Request is sent to the Presentation Layer.
3. Business Logic Layer processes the request.
4. Data Access Layer retrieves data from databases.

5. Business Logic Layer analyses data.
6. Dashboard displays charts and KPIs.
7. Reports are generated.
8. User downloads reports.

4.7 Overall Workflow

- User Login
- User Authentication
- Data Collection
- Data Processing
- KPI Calculation
- Dashboard Visualization
- Report Generation
- Decision Making
- Data Backup

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5. Quality Attributes

Scalability

The layered architecture supports scalability by allowing each layer to be upgraded independently. Additional servers, databases, and cloud resources can be added as business requirements increase.

Security

Security is achieved through authentication, authorization, encrypted communication, secure APIs, role-based access control, and regular database backups. Only authorized users can access sensitive business information.

Performance

Performance is improved through optimized database queries, data caching, efficient business logic processing, and responsive dashboards. Real-time data visualization ensures quick access to business information.

Reliability

The system ensures reliable operation through automatic backup, fault tolerance, error handling, data validation, and continuous monitoring. This minimizes system failures and protects business data.

Maintainability

The modular layered architecture simplifies software maintenance. Developers can update individual components without affecting the entire system, reducing maintenance time and improving software quality.

Availability

The dashboard is designed for high availability through cloud deployment, backup servers, load balancing, and disaster recovery mechanisms, ensuring continuous access for business users.

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6. Justify Architectural Decisions

Rationale

The Layered Architecture was selected because it provides a clear separation of responsibilities among different system components. It simplifies development, testing, maintenance, and future enhancements.

The architecture also supports integration with external enterprise systems, making it highly suitable for Business Intelligence applications.

Trade-offs

Advantages

- Easy to develop.
- Easy to maintain.
- Highly secure.
- Modular design.
- Supports scalability.
- Easy integration with external systems.
- Better code reusability.

Disadvantages

- Additional communication between layers.
- Slight increase in processing time.
- More components compared to simple architectures.
- Initial development effort is higher.

Future Enhancements

The selected architecture supports future enhancements such as Artificial Intelligence-based predictive analytics, Machine Learning models for business forecasting, cloud-native deployment, mobile dashboard applications, chatbot integration, real-time streaming analytics, and IoT data integration. Since the architecture is modular, these features can be incorporated without significant changes to the existing system. This ensures long-term maintainability, flexibility, and adaptability to future business needs.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all major components, interfaces, and interactions accurately represented.	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and lacks sufficient detail.	Diagram is incomplete, unclear, or technically incorrect.
Component Interaction and Quality Attribute Analysis (15%)	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation (10%)	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.

	organized, professional report.			

Criteria	Mark
System Requirement Analysis **(30%)**	/5
Selection of Software Architecture **(25%)**	/5
Architecture Diagram and Component Design **(20%)**	/5
Component Interaction and Quality Attribute Analysis **(15%)**	/5
Architectural Justification and Technical Presentation **(10%)**	/5
TOTAL	/25